Elastic response of aluminium to magnetostriction of an $Al_{13}Fe_4$ inclusion and the extent of possible plasticity

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Abstract

Molecular dynamics at 0 K gives stresses around a spherical $Al_{13}Fe_4$ inclusion of radius 2.5 nm in aluminium, lengthened by magnetostriction along the magnetic field and shortened across it. The matrix is compressed by 126 MPa at the ends of the inclusion and stretched by 75 MPa at its sides; shear stresses reach 28 MPa. The response is elastic: tenfold strain forms no dislocations, and a dislocation above the inclusion shifts 0.78 Å and returns on unloading. A compliant interface layer makes the atomistic stresses about 80% of the elastic solution. Around a micron-size particle the elastic stress exceeds a yield stress of 40–60 MPa within about one radius, and one of 120–151 MPa only in a thin surface layer. With the activation volume measured for Al–Mg–Si, the observed creep increase would need 6–7 times the measured Fe–Al magnetostriction. Why the effect outlasts the field remains unexplained.

Keywords molecular dynamics; magnetostriction; inclusions; internal stress; aluminium alloys

1 Introduction

Al–Mg–Si alloys containing iron-rich inclusions creep faster after they have been held in a static magnetic field (creep is slow deformation under a constant load), and the effect persists after the field is removed [1–4]. After holding in a 0.7 T field the creep rate rises by up to 25% and the microhardness falls by up to 30% [1]. The extra strain of the samples grows linearly with the induction between 0.2 and 0.7 T [3], and the heat released in the subsequent deformation grows fivefold [5]. The inclusions are 1–5 μm in size [1, 5].

One proposed explanation is magnetostriction of the inclusions, the change of their shape in the field. The matrix resists the change, and stresses appear around the inclusion that can move dislocations, the line defects whose motion carries plastic deformation. The torque with which the field turns a magnetised particle gives only about 0.1 MPa at the interface [1]. The same papers estimate the magnetostrictive stress as 147 MPa [3], 560 MPa [1] and 650 MPa [6]. Each estimate is a single number: it does not say how the stress is distributed around the particle, how fast it falls with distance, or how much of it acts on dislocations.

Here the stress field around one $Al_{13}Fe_4$ inclusion in aluminium is calculated by molecular dynamics, which follows the atoms under the forces between them. The forces come from an interatomic potential (a formula for the energy of interacting atoms). The potential does not describe magnetism, so the strain of the inclusion is prescribed; its magnitude corresponds to the 147 MPa estimate. The atomistic result is compared with the elastic solution, and the field is then used to estimate the force on dislocations and the increase in creep rate.

The question addressed here is whether the magnetostriction of an inclusion can by itself deform aluminium plastically, that is, create dislocations or move them irreversibly, and if so, in what zone around the particle. The calculation shows that at the given strain the response of aluminium is elastic, and plastic flow is possible only in a layer next to the particle whose thickness depends on the yield stress of the alloy.

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2 Model and method

2.1 The cell

The cell is a periodic box of aluminium, $12.6 \times 12.4 \times 12.6$ nm. A spherical $Al_{13}Fe_4$ inclusion of radius $R = 2.5$ nm sits at its centre (Fig. 1a). The inclusion is cut from an $Al_{13}Fe_4$ crystal with the lattice of the crystal-structure data [7]; aluminium atoms closer than 2.2 Å to an inclusion atom are removed. The cell holds 118 861 atoms, 114 374 in the matrix and 4 487 in the inclusion. Aluminium surrounds the inclusion on all sides. The boundary conditions are periodic along all three axes, so the cell represents a regular array of inclusions 12.6 nm apart along x and z and 12.4 nm along y; their volume fraction is 3.3%.

A sphere was chosen as the simplest compact shape; the actual shape of the inclusions is not reproduced. The stress around a sphere falls with the distance r from its centre as r^{-3} and is small two or three radii away, so a cell of 10^5 atoms is large enough. A sphere also has an exact elastic solution [8, 9] against which the atomistic result can be checked.

The cell axes are $x \parallel [1\bar{1}0]$, $y \parallel [11\bar{2}]$ and $z \parallel [111]$. The (111) planes on which dislocations glide in aluminium are then horizontal, and the slip direction $[1\bar{1}0]$ lies along x. The interatomic potential for Al–Si–Mg–Cu–Fe is taken from Ref. [10] (modified embedded-atom method); the calculations use LAMMPS [11]. The elastic constants of aluminium with this potential, $C_{11} = 110.5$, $C_{12} = 60.9$ and $C_{44} = 28.4$ GPa, are close to the measured ones.

2.2 The given magnetostrictive strain

The magnetic field points along x. Magnetostriction lengthens the inclusion along the field and shortens it across, without changing its volume (Fig. 1b):

$$\varepsilon_{xx} = \varepsilon, \qquad \varepsilon_{yy} = \varepsilon_{zz} = -\varepsilon/2, \qquad \varepsilon = 1.94 \times 10^{-3}. \tag{1}$$

This ε comes from the 147 MPa estimate of Ref. [3]. That estimate is the Young's modulus of aluminium, 70 GPa, times the extra strain that the samples accumulate in creep after holding in a 0.7 T field (about 0.21%); the Young's modulus is the stress needed per unit strain. Here 147 MPa is divided by the Young's modulus measured on the alloy samples, 75.7 GPa [5]. The magnetostriction of the $Al_{13}Fe_4$ inclusions themselves has not been measured, so ε is an input of the calculation. Every stress below is proportional to ε; for another strain ε' it is multiplied by $\varepsilon'/0.194\%$. With the same modulus, the estimates of 560 and 650 MPa [1, 6] correspond to strains of 0.74–0.86%, 3.8 and 4.4 times the given one; the calculations at 4ε and 10ε (Supplementary S2) cover this range.

Each inclusion atom is moved by $\mathbf{u} = \boldsymbol{\varepsilon}(\mathbf{r} - \mathbf{r}_c)$, where $\mathbf{r}$ is the position of the atom before the move and $\mathbf{r}_c$ the inclusion centre, and is then held fixed. The aluminium atoms move until the forces on them vanish (energy minimisation). This gives the inclusion the shape it is assumed to take under the field; in a real inclusion magnetic forces would maintain it, if a magnetostriction of this size exists. The arrows in Fig. 1b show the forces with which the inclusion then acts on the aluminium.

The strain (1) is the strain of the inclusion inside the matrix. A free particle would lengthen more in the same field, because no matrix would hold it back. If 0.194% were the free magnetostriction of a particle two to two and a half times stiffer than aluminium, every stress would be about one third smaller than computed (Supplementary S3).

The reference cell without strain is prepared once (heating to 300 K, cooling and energy minimisation; Supplementary S1). Its inclusion atoms are fixed, because with this potential the equilibrium lattice of $Al_{13}Fe_4$ is 8% larger in volume than the measured one and a free inclusion would swell. The cells with strain $\pm\varepsilon$ are built from it by moving the inclusion atoms at fixed cell size; comparing the responses to $+\varepsilon$ and $-\varepsilon$ shows how far the response departs from linear (Supplementary S2).

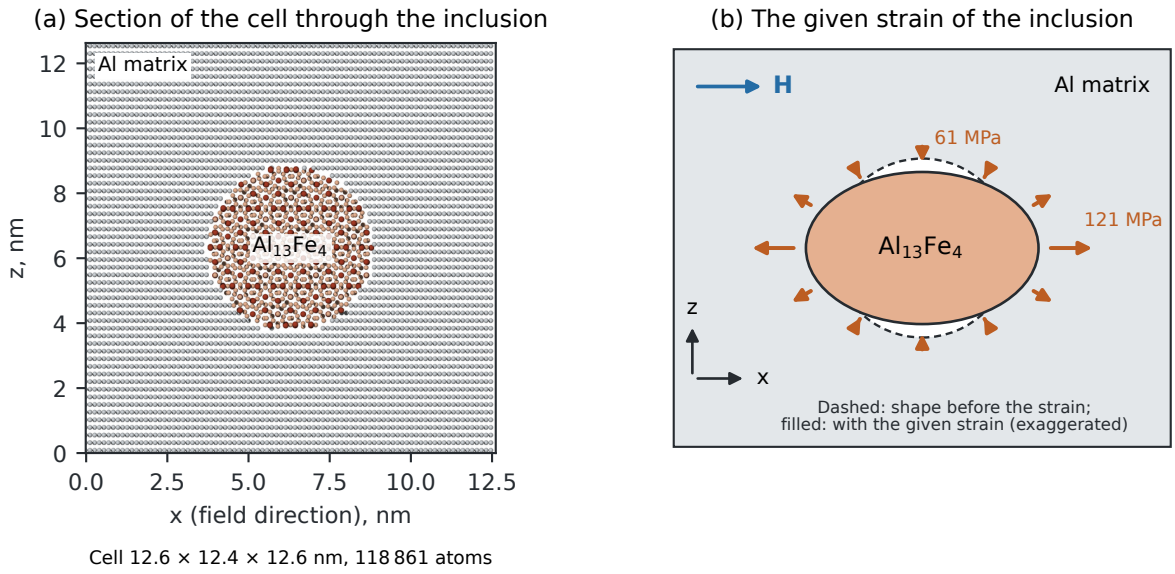


Figure 1: The model. (a) Section of the cell by the xz plane through the inclusion centre, a 1 nm thick slab (OVITO [12]); grey dots are aluminium atoms of the matrix, orange and red ones Al and Fe atoms of the inclusion. (b) The given strain of the inclusion (exaggerated): dashed circle, original shape; filled, shape with the strain (1). Arrows: forces with which the inclusion acts on the aluminium in the single-sphere elastic solution, outward at the ends along the field and inward on top and at the bottom; arrow length is proportional to force per unit surface area, and the numbers give this force along and across the field.

2.3 How the stresses are obtained

The stress at each matrix atom is computed from the forces of its neighbours (the virial of the atom) and divided by the atomic volume of aluminium, 16.61 Å^3; in a uniform crystal the mean of these atomic stresses is the ordinary stress. The quantity used is the difference between the strained cell and the reference cell, that is, the stress added by the magnetostriction. Tension is positive and compression negative. Three quantities are used. The normal stress along the radius, σ_{rr}, shows how hard the matrix pushes on the inclusion or pulls it; the estimates of 147–650 MPa are of this kind. The equivalent (von Mises) stress, σ_{eq}, combines all components except the hydrostatic pressure into one number that is compared with the yield stress, the stress at which the alloy starts to deform plastically. The shear stress $\tau = \sigma_{xz}$ shears the crystal along the horizontal slip plane in the x direction and so pushes an edge dislocation with Burgers vector along x. The Burgers vector is the shift of the crystal left behind when the dislocation passes; for an edge dislocation it is perpendicular to the line.

Values are averaged over atoms in spherical shells 2 Å thick inside narrow cones about chosen directions: along the field, across it, and at 25° and 67° to the field, where the shear stress is largest (within 10° of the direction, 7.5° for the inclined ones). The distance $d = r - R$ is measured from the inclusion surface. For the maps in Fig. 2 the stresses are averaged over rotations about the field axis: the field is symmetric about it apart from the weak dependence of the elasticity of aluminium on direction.

2.4 The elastic solution

Elasticity theory treats the metal as a continuous elastic medium without atoms. The atomistic result is compared with two of its solutions. The first is exact: a single sphere with the same strain in an infinite isotropic matrix [8, 9, 13]. The second is computed numerically for the same

periodic cell [14] with the cubic elastic constants of aluminium from the potential. It includes the neighbouring inclusions and the fixed cell size, so it can be compared with the atomistic result directly. The first shows the field of a single inclusion in a large grain and carries over to particles of real size.

3 Results

3.1 How the stress is distributed

Fig. 2 shows how the added stress is distributed around the inclusion. Along the field, at the ends of the inclusion, the matrix is compressed: the inclusion pushes it apart. Across the field the matrix is in tension: the inclusion draws back from it. The shear stress on the slip planes vanishes on the x and z axes and is largest in four inclined lobes, at about 25° and 67° to the field, changing sign from lobe to lobe. The atomistic map and the elastic map for the same cell agree in shape, but the atomistic stresses are about one fifth smaller (Section 3.2). Within 12 Å of the interface the shape differs as well, because a continuum does not describe the atomic structure of the boundary.

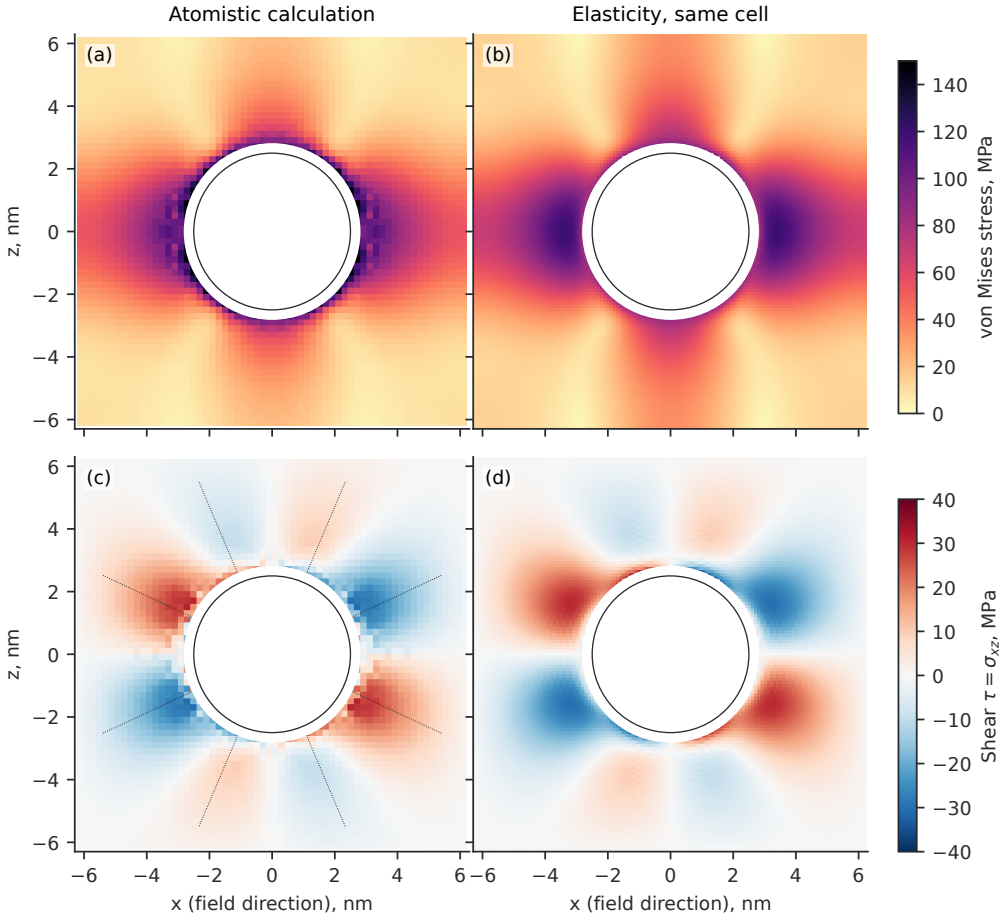


Figure 2: Stresses added by the strain of the inclusion in the xz plane through its centre: equivalent (von Mises) stress (top) and shear stress on the slip plane, $\tau = \sigma_{xz}$ (bottom). Left, atomistic calculation; right, elastic solution for the same periodic cell. The field points along x; the circle is the inclusion surface, and the white ring around it (3 Å of matrix) is left blank. Dotted lines in the lower left panel: directions at 25° and 67° to the field.

3.2 How the stress falls with distance

Fig. 3 gives the profiles along the chosen directions. The atomistic values quoted here are for shells 5–10 Å from the surface; closer in, the narrow cones along the field hold only 15–20 atoms per shell, and the values scatter by up to 25 MPa. Along the field the normal stress is compressive, about 126 MPa, which is close to the 147 MPa estimate of Ref. [3]. The agreement is expected, because ε was derived from that estimate. Across the field the normal stress is tensile, about 75 MPa. At the sides of the inclusion, near the plane $x = 0$, the stress along the field is also large: as the inclusion lengthens, it pulls the aluminium next to it along, and σ_{xx} right at the surface there reaches 121 MPa. The shear stress on the slip planes reaches 28 MPa in the lobes.

One radius from the surface ($r = 2R$) the normal stress along the field falls to about 54 MPa and the shear stress in the lobes to about 19 MPa; two radii away ($r = 3R$) they are about 18 and 7 MPa (single-sphere solution).

From $d \approx 10$ Å on, the atomistic stresses equal the elastic solution for the same cell times 0.80 (0.80 to 0.94 for the different directions); after this scaling they differ from it by 2.5% on average. The atomic displacements, which do not depend on how the stress is computed, are reduced in the same way. The matrix therefore deforms less than with a perfectly bonded interface. The deficit builds up in a layer 5–8 Å thick at the interface: the aluminium lattice there is distorted, the first layer has half the bulk density, and the layer passes on only part of the strain of the inclusion (Supplementary S2).

If the compliance of the layer does not depend on particle size, the weakening scales inversely with the radius and would be about 0.02% for a particle of radius 2.5 μm, so the elastic solution is used below for real particles. This is untested: one inclusion radius and one interface were computed, at 0 K, and sliding at the interface at room temperature would weaken the field whatever the particle size.

The far field in Fig. 3d is shown for a single sphere: the volume fraction of inclusions in the alloy, about 0.25%, is more than ten times smaller than in the cell (Supplementary S4 compares the two solutions). The field of a single sphere depends only on r/R, so for a particle 5 μm across the distances in Fig. 3 are simply multiplied by a thousand (upper scale).

The experimental papers give different yield stresses for the alloy: 40–60 MPa in Ref. [3] (a handbook value for commercially pure aluminium; the samples of that work began to deform plastically at about 42 MPa) and 120–151 MPa in Refs. [1, 5], where 151 MPa was measured on samples after the creep tests. For 40–60 MPa the equivalent stress exceeds the yield stress out to $r \approx 1.9$–$2.2R$, in a shell about one radius thick (2.2–3.1 μm for a 5 μm particle), and for 120–151 MPa only out to $r \approx 1.02$–$1.06R$. Slip starts when the largest shear stress on the twelve slip systems (a slip plane together with a slip direction in it) exceeds the critical resolved shear stress, the yield stress divided by the Taylor factor 3.06, which accounts for the different orientations of the grains in a polycrystal [15]. A critical stress of 13–20 MPa is exceeded out to $r \approx 2.2$–$2.6R$, and one of 39–49 MPa out to $r \approx 1.1$–$1.5R$. The plastic zone of about 6 μm in Ref. [1] assumes that the stress falls as $1/r$; with the r^{-3} decay the same estimate gives about 2 μm. Where the stress exceeds the yield stress, the elastic solution overestimates it: dislocations already present in the alloy would start to move and part of the stress would relax. The atomistic calculation does not test this, because its matrix has no pre-existing dislocations, and at 0 K no new ones form even at a lobe shear stress of 306 MPa (strain 10ε, Supplementary S2).

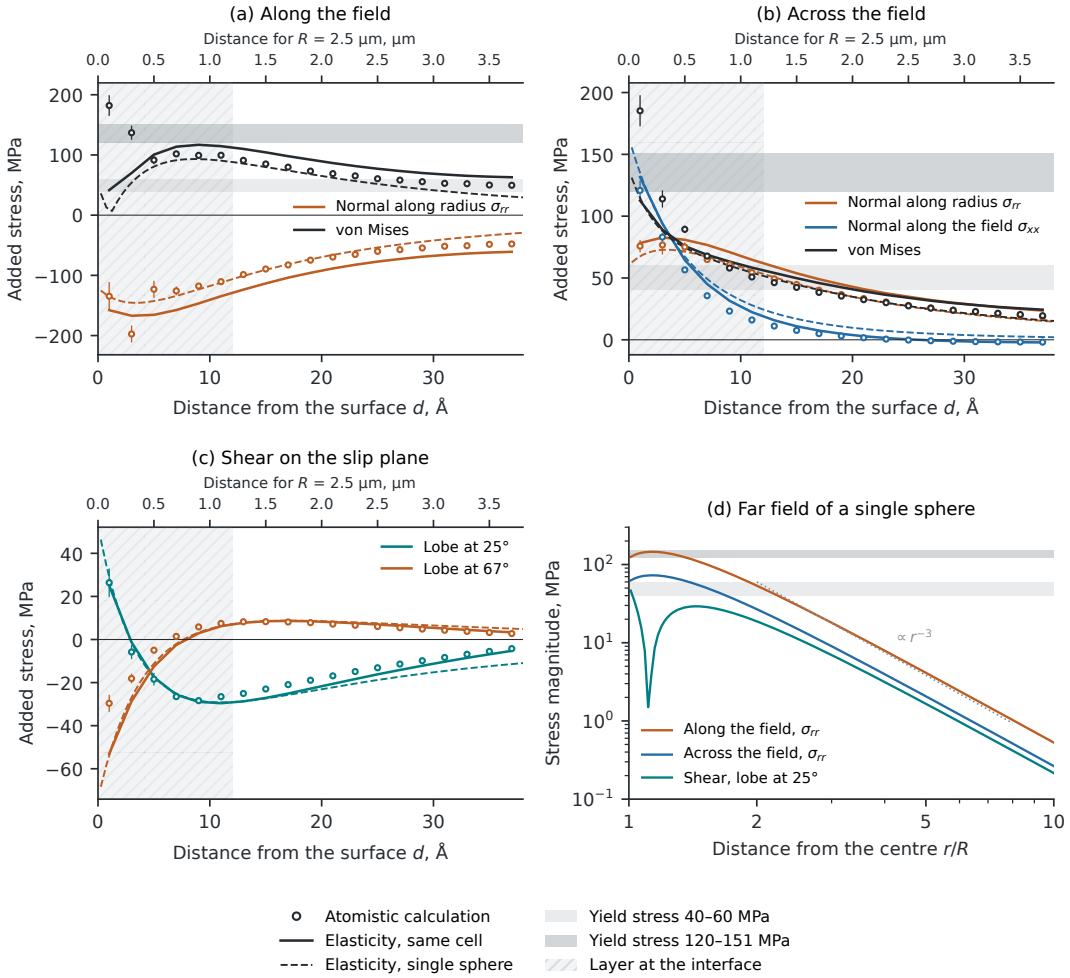


Figure 3: Added stresses against the distance d from the inclusion surface: (a) along the field, (b) across the field, (c) shear stress in the 25° and 67° lobes (symbols and lines as in the legend below; symbols are means over 2 Å shells with their errors). In the hatched layer at the interface the atomistic field differs in shape from the elastic one; farther out it is about 80% of the same-cell solution. The yield-stress bands are compared with the von Mises stress. Upper scale: the same distance for a particle of radius 2.5 μm. (d) Magnitude of the single-sphere stress on logarithmic scales; dotted, the slope r^{-3}; the dip of the shear stress near $r \approx 1.1R$ is where it changes sign.

3.3 The force on dislocations

A shear stress τ exerts on a dislocation with Burgers vector $b = 2.86$ Å a force τb per unit length, directed along the slip plane (the Peach–Koehler force [16]). The dislocation moves if this force exceeds what holds it back: the lattice resistance (the Peierls stress, the least stress at which a straight dislocation in a pure crystal at 0 K moves to the next position), solute atoms and other dislocations.

Fig. 4 shows the shear stress of the inclusion field averaged along a straight line on which an edge dislocation can lie. The line runs parallel to y at a height h above the inclusion surface. The stress changes sign when x is replaced by $-x$: on one side of the inclusion it pushes the dislocation forward, on the other side back. The averaged stress is largest, 5.8 MPa, at $h = 10$ Å. This is below the Peierls stress of the potential, about 9 MPa (dashed lines in Fig. 4). On the short part of the line right above the inclusion the stress is larger, up to 12.2 MPa, and this part could bow out even when the average is too small to move the whole line. The line average depends on the spacing of the inclusions in the model (12.4 nm along the line) and does not

carry over directly to the alloy.

In an Al–Mg–Si alloy dislocations are also held by Mg and Si atoms and by precipitates (fine particles of Mg–Si compounds inside the grains), so the resistance to their motion is the critical resolved shear stress, 13–49 MPa. The field of the inclusion exceeds it only next to the particle (Section 3.2). Farther out it cannot by itself push dislocations past their obstacles and can only speed up or slow down their thermally activated motion, in which thermal vibrations of the atoms carry them past the obstacles (see the creep estimate below).

Whether the field moves a dislocation was checked directly (Fig. 5). An aluminium slab contains the same held inclusion and a straight edge dislocation whose line runs through the whole slab along y, on the slip plane 12 Å above the inclusion (Supplementary S6). In aluminium such a dislocation splits into two partial dislocations with a ribbon about 9.2 Å wide between them, where the stacking of the atomic planes is faulted. With the inclusion unstrained, the dislocation lies, after energy minimisation, 20.4 Å from the inclusion centre along x, where the line-averaged shear stress of the field is about 4.6 MPa, about half the Peierls stress. When the inclusion is given the strain (1), the whole dislocation line moves 0.78 Å away from it on the slip plane. The dislocation moves through the lattice in steps of $b/2 = 1.43$ Å, and each step requires crossing an energy barrier. The shift is less than one step, and the dislocation does not take the step. When the strain is removed, the dislocation returns to within 0.06 Å of where it was: in this calculation at 0 K the shift is elastic and reversible. At room temperature the barriers are easier to cross, and the same field could move the dislocation further.

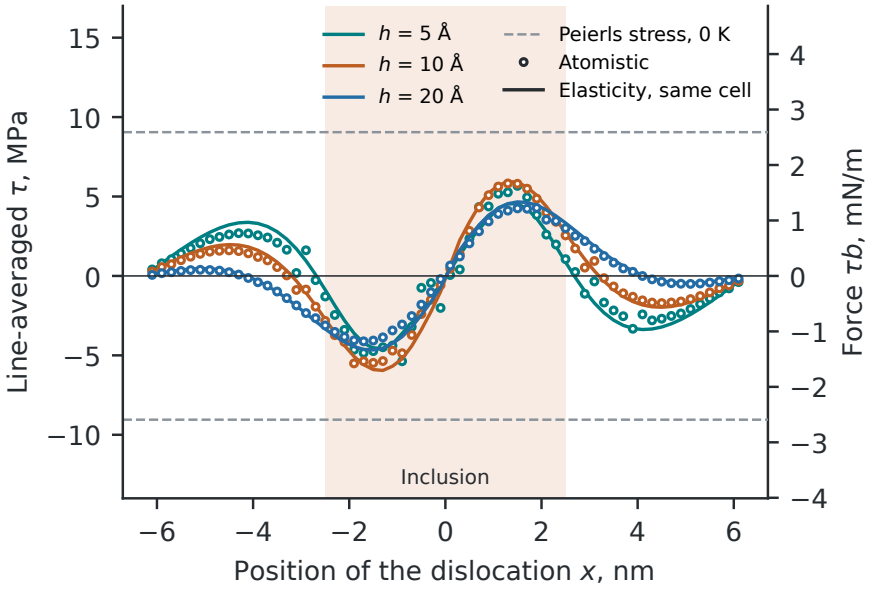


Figure 4: Shear stress averaged along a straight edge dislocation 12.4 nm long that runs above the inclusion at a height h above its surface, against the position x of the dislocation. Symbols, atomistic calculation; lines, elastic solution for the same cell; shaded, the extent of the inclusion along x; dashed lines, the Peierls stress of a straight edge dislocation with this potential at 0 K. Right scale: force per unit length of the dislocation, τb.

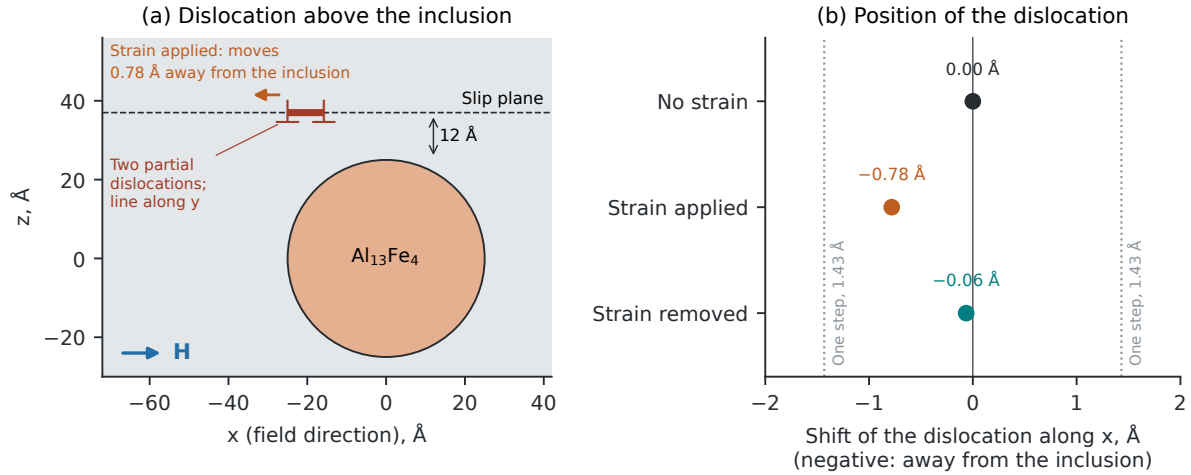


Figure 5: The edge dislocation above the inclusion. (a) Section of the slab by the xz plane through the inclusion centre, to scale: the inclusion, the slip plane 12 Å above it and the two partial dislocations ($\perp$ symbols) with the faulted stacking between them; the dislocation line is normal to the figure. The arrow shows where the dislocation moves when the inclusion is strained. (b) Shift of the dislocation (the midpoint of the partials) along x from its position without strain. Dotted, the positions of the dislocation one lattice step away ($b/2 = 1.43$ Å); an irreversible shift would mean passing into one of them.

3.4 An estimate for creep

The link between the stress field and the creep rate is estimated with a model of thermally activated glide [17, 18]. The estimate applies to creep while the field is on and the stresses around the inclusions exist; the observed increase was measured after the field was removed (see the Discussion). A dislocation passes its obstacles with the help of thermal fluctuations. A shear stress τ makes forward jumps $\exp(V^*\tau/k_\mathrm{B}T)$ times more likely and backward jumps less likely by the same factor; V^* is the activation volume, a measure of how strongly the barrier depends on stress, and $k_\mathrm{B}T$ is the thermal energy. If dislocations around an inclusion meet stresses of both signs equally often, their mean velocity grows by the factor $\cosh(V^*\tau/k_\mathrm{B}T)$, the mean of the factors $\exp(V^*\tau/k_\mathrm{B}T)$ and $\exp(-V^*\tau/k_\mathrm{B}T)$ for the two signs. The excess of this factor over one, integrated over the matrix around one inclusion and multiplied by the number of inclusions per unit volume, f/V_p, gives the relative increase in creep rate (derived in Supplementary S5),

$$H = \frac{f}{V_p} \int_{r>R} \left\langle \cosh \frac{V^* \tau_s(\mathbf{r})}{k_\mathrm{B}T} - 1 \right\rangle \mathrm{d}V, \tag{2}$$

where f is the volume fraction of inclusions, V_p the volume of one inclusion, τ_s the shear stress on slip system s, and the angle brackets an average over the twelve slip systems and over grain orientations. The elastic field of a single sphere goes into the integral (the atomistic weakening is taken as negligible for micron-size particles, Section 3.2); it depends only on r/R, so H does not depend on particle size. The values used are $T = 300$ K and $f = 0.246\%$; the latter corresponds to 0.35 wt% of inclusions [3].

In Fig. 6 the solid curve shows the amplitude τ_m needed for the 25% increase in creep rate observed after holding in the field [1] at different V^* if the shear stress falls from the inclusion surface as $\tau_m(R/r)^3$; this amplitude falls as the activation volume grows. The dashed curve shows the amplitude τ_m for which such a field gives the same increase as the computed field (Supplementary S5). The curves cross at $V^* \approx 23\,b^3$. At $V^* = 70\,b^3$, measured for an Al–Mg–Si alloy [19], the computed field would raise the rate 10^8 times, which is outside the range of the model. Because H depends only on the product $V^*\varepsilon$, at $V^* = 70\,b^3$ an inclusion strain of about

0.065% is needed. The measured magnetostriction of Fe–Al alloys does not exceed 0.01% [20, 21], six to seven times less.

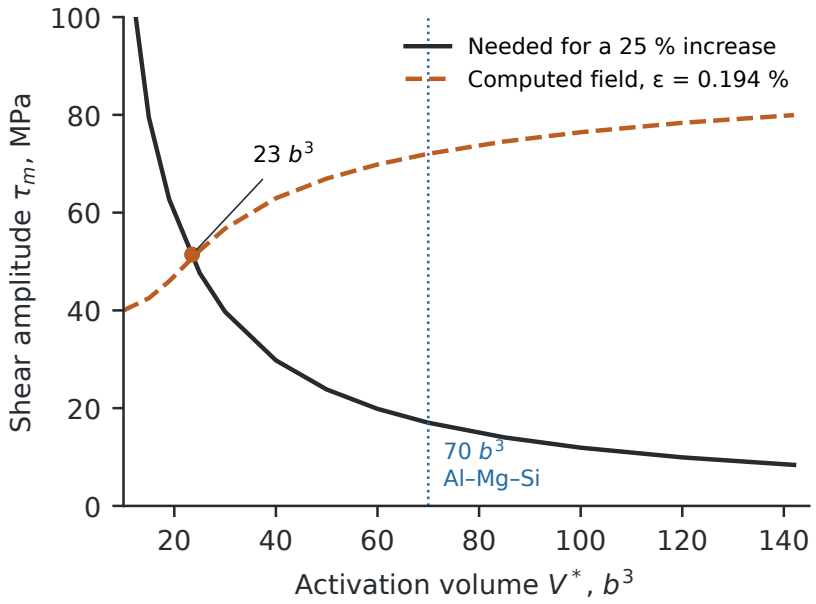


Figure 6: Shear-stress amplitude needed for a 25% increase in creep rate (solid line) and the amplitude equivalent to the elastic single-sphere field of the inclusion with a strain of 0.194% (dashed), against the activation volume V^*; $T = 300$ K, $f = 0.246\%$. Dotted line: $V^* = 70\,b^3$ for Al–Mg–Si [19].

4 Discussion

The calculation gives the stresses that an inclusion of a given shape produces in aluminium. The magnetic side remains open: the magnetostriction of the inclusions themselves has not been measured. Alloys with these inclusions show magnetic hysteresis [1, 3, 5], that is, their magnetisation depends on the field history. According to Ref. [1] the particles are superparamagnetic above about 250 K: they are small enough for thermal fluctuations to flip their magnetisation. The phase of the inclusions is identified differently: X-ray diffraction points to $Al_{13}Fe_4$ [3, 5], while electron microscopy and Mössbauer spectroscopy point to FeAl and Fe_3Al alloys [1]. Strong ferromagnetism has not been established for bulk $Al_{13}Fe_4$ [22, 23], and the magnetostriction of Fe–Al alloys does not exceed 0.01% [20, 21].

If the real strain of the inclusions is close to the magnetostriction of Fe–Al alloys, the stresses are twenty times smaller than computed and do not reach the yield stress. In the single-sphere solution plastic flow at the particle surface becomes possible once the inclusion strain exceeds 0.028–0.067% in the soft alloy and 0.084–0.168% in the stronger one (the lower value from the shear stress, the upper from von Mises). For the free magnetostriction of an elastic particle these values are about one and a half times larger (Section 2.2). If the estimates of 560 and 650 MPa hold (a strain of 0.74–0.86%), the equivalent stress would exceed even a yield stress of 120–151 MPa out to $r \approx 2.2$–$2.6R$.

The calculation is static (energy minimisation without thermal motion of the atoms) and done at 0 K for inclusions of one size in a regular array. The inclusion atoms are held at the measured lattice, so the elasticity of the inclusion itself does not enter. The matrix is pure aluminium without Mg and Si atoms, precipitates or pre-existing dislocations, so the yield stress is taken from experiments [1, 3, 5] and plastic flow near the inclusion is not modelled.

The main limitation concerns the persistence of the effect. In the experiments creep is measured after the field is removed, whereas the elastic stress of magnetostriction disappears with the field. If the field moves dislocations near the inclusions, the displaced dislocations may stay in their new positions after the field is removed; the computed shell where the stress exceeds the yield stress shows where to look for such changes. In the one case computed, a straight dislocation at 0 K, the dislocation returned when the strain was removed; the calculation does not show whether such a rearrangement of defects persists in the alloy or how it affects subsequent creep.

5 Conclusions

1. At the given magnetostrictive strain of 0.194% the aluminium matrix around an $Al_{13}Fe_4$ inclusion deforms elastically: at 0 K no dislocations form even at ten times the strain, and a dislocation above the inclusion shifts by 0.78 Å and returns when the strain is removed.

2. A compliant aluminium layer at the interface makes the atomistic field about 80% of the elastic solution; for micron-size particles this weakening is probably negligible, but this is untested.

3. Plastic flow is possible only in a zone next to the particle where the elastic stress exceeds the yield stress: a shell about one radius thick for a yield stress of 40–60 MPa and a thin surface layer for 120–151 MPa.

4. With the measured activation volume of $70\,b^3$ the observed increase in creep rate would need a strain of 0.065%, six to seven times the measured magnetostriction of Fe–Al alloys. The calculation does not explain why the effect persists after the field is removed.

Author contributions

I. Mikhailovskiy developed the method and software, ran the calculations, analysed the results, made the figures and wrote the text. D. Pshonkin conceived the study, supervised it, and reviewed and edited the text.

Data and code availability

The LAMMPS inputs, the programs that build the cell, analyse the results and draw the figures, the reference and strained configurations with atomic stresses, and the result records are available at `https://github.com/iluuua/magnetostriction-plasticity-bounds`.

Conflict of interest

The authors declare no conflict of interest.

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